

average g-force of the plane during the simulated landing with the control subjects who had undergone brain-stimulation training.

A tDCS signal can boost the brain and heal neural impairments. However, this study takes a new direction in the field of artificial intelligence and neural systems. HRL's findings are the first to ensure the fact that a tDCS signal can help improve practical learning.

Tech that lets smartwatches track fingers in mid-air

Scientists, including those of Indian origin, have developed a new sonar technology that allows you to interact with mobile devices by writing or gesturing on a tabletop, a sheet of paper or even in mid-air.



FingerIO allows users to interact with mobile devices by writing or gesturing on any nearby surface by turning a smartphone or smartwatch into an active sonar device (Image courtesy: www.newseveryday.com)

An app, FingerIO, tracks fine-grained finger movements by turning a smartphone or smartwatch into an active sonar system using the device's own microphones and speakers. Since sound waves travel through fabric and do not require a line of sight, you can even interact with a phone inside a front pocket or a smartwatch hidden under a sweater sleeve.

Researchers at University of Washington, USA, showed that FingerIO can accurately track two-dimensional finger movements within 8mm, which is sufficiently accurate to interact with today's mobile devices.

Technology that uses smartphone to count molecules

Diagnostic healthcare is many a time restricted in areas with limited resources because the procedures required to detect many molecular markers that can diagnose diseases are too complex or expensive to be used outside of a central laboratory. To address the need for a robust readout system for quantitative diagnostics, researchers from Rustem Ismagilov's lab have invented a new visual readout method that uses analytical chemistries and image processing to pro-

**Now BINAY
White-Tunable
Vario-White
Circadian
PowerLED
Lights
can enhance
your health**

BINAY Human-Centric Circadian Vario-White PowerLED Lights With Adjustable Colour Temperature are designed to affect the human circadian rhythm positively for enhanced physical and mental well-being

Binay Opto Electronics Pvt. Ltd.
27, PANDITIA TERRACE, Calcutta 700 029, India
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
Email: info@binayled.com, Fax: 91-33-2242 1493

www.binayled.com

BINAY Multichip LED Lamps

High reliability direct-retrofit LED bulbs for Control and Instrumentation applications

Ba9s bayonet pin cap	E-10 screw cap
Midget Flange cap	Midget Groove cap
5530 and 6844 slide base	Wedge base

Simple retrofits for existing filament bulb holders, in applications such as annunciator window (fascia) backlighting and illuminated pushbutton switches, where the reliability of indication is important or where lamp replacement is difficult or expensive

- Long life (typically 100,000 hours)
- Low power consumption; Low heat generation
- High brightness and wide-angle light emission
- Shock-proof and vibration resistant
- High reliability of indication
- One-time Expense

Binay Opto Electronics Pvt. Ltd.
27, PANDITIA TERRACE, Calcutta 700 029, India
Telephone: +91 (33) 4006 9875, 2475 0392, 2475 0030
email : info@binayled.com, <http://www.binayled.com>

POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983